Deep Isolation Forest Outlier Analysis of Large Multimodal Adolescent Neuroimaging Data

UC San Diego
JACOBS SCHOOL OF ENGINEERING

E. Silberman, M. Wagner, A. Camassa, T. Sejnowski
University of California San Diego, Salk Institute of Biological Studies

salk Institute for Biological Studies

Introduction

Outliers in neuroimaging data offer crucial insights about **data quality, acquisition variability**, and **potential confounding factors** that may impact analysis

ABCD Study: Multimodal data from **~12,000 subjects**, providing a **large dataset** to identify outliers

We applied **Deep Isolation Forest (DIF)** to detect outliers from resting-state functional connectivity data

Used tuned **L1 Logistic Regression** to **predict outliers**





Pipeline

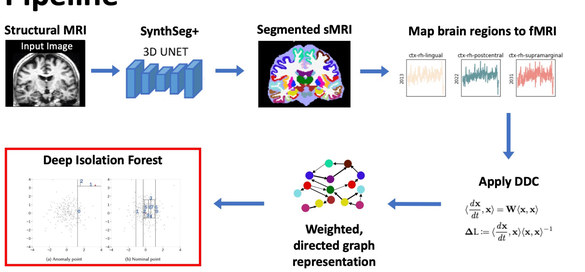


Fig. 1 Data processing pipeline for anomaly detection in resting-state functional connectivity

Dynamical Differential Covariance (DDC) & Deep Isolation Forest

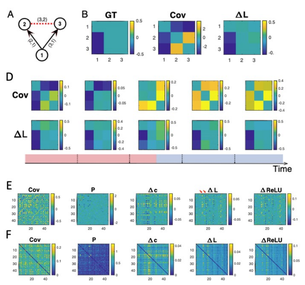


Fig. 3 From Y. Chen et al. (2022). **A** simulated network and its ground truth connections; **B** comparison between DDC and Cov. **E** Simulation of a two- states dynamical system; **F** Sample covariance estimation shifted between states while ΔL estimation consistently reported the static true connectivity.

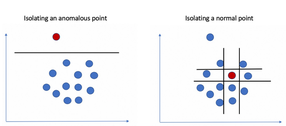


Fig. 4 From F. T. Liu et al. (2008) Isolation Forest Diagram

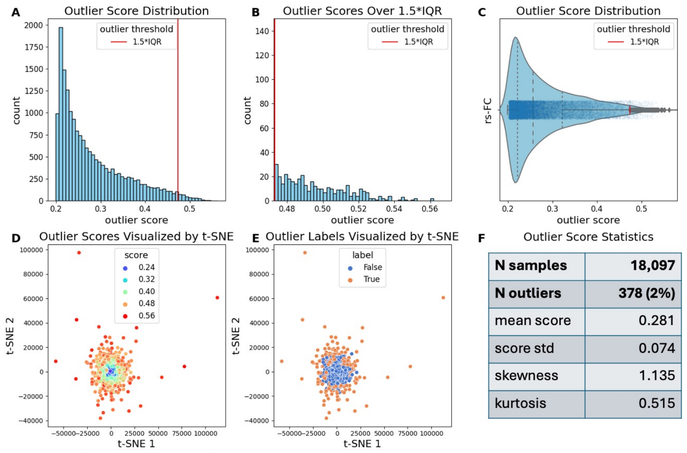
Isolation Forest: Partitions data to isolate outliers quickly.

Deep Learning Extension: Captures complex, non-linear relationships.

Anomaly Score: Measures how easily a point is isolated — higher score = outlier.

Advantages: Handles high-dimensional data, robust to noise, models non-linear patterns.

Results



N samples	**18,097**
N outliers	**378 (2%)**
mean score	0.281
score std	0.074
skewness	1.135
kurtosis	0.515

Fig. 5 Used Xu et al. (2022). Deep Isolation Forest
A outlier scores distribution with threshold; **B** outlier scores distribution past threshold; **C** outlier score distribution; **D** outlier scores visualized; **E** outlier labels visualized; **F** outlier score statistic

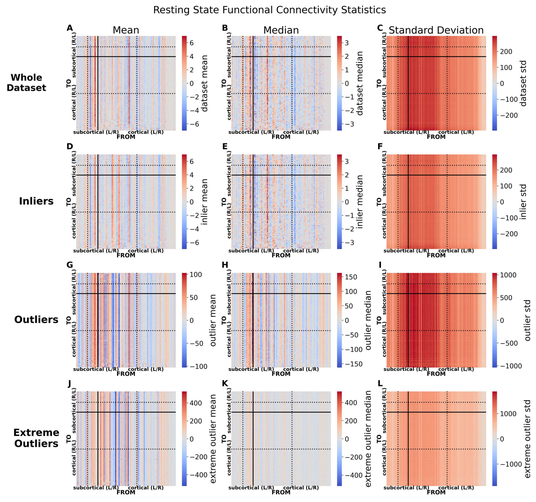
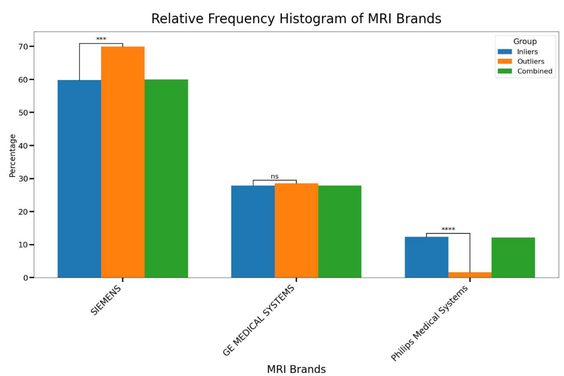
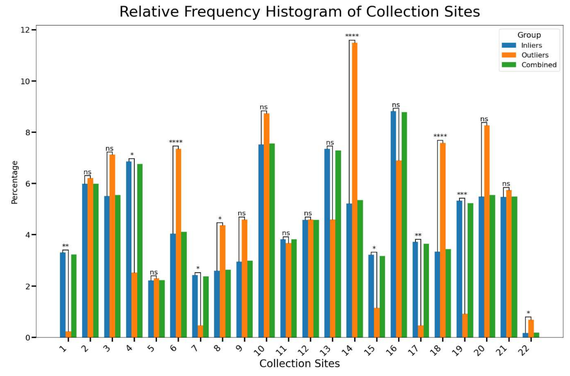


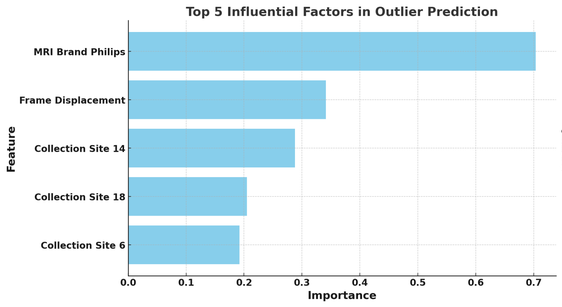
Fig. 6 Heat maps showing mean, median, and std of functional connectivity for whole dataset, inliers, outliers, and extreme outliers

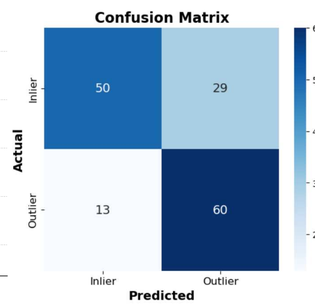
Statistical Differences in Outliers vs Inliers

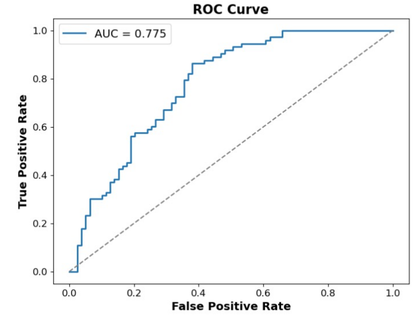




Prediction of Outliers using Logistic Regression







Discussion

Outlier Prediction Result:
AUC = 0.775

MRI Brand (Philips) was the strongest predictor, but Philips scanners had the fewest outliers

Motion (Frame Displacement) strongly influenced outlier classification

Collection sites greatly influenced outlier status

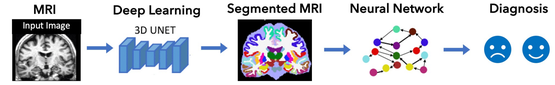
Future Direction

Determine if **Philips scanners produce inherently better data** or if **site-level differences** explain lower outlier rates

Re-evaluating if **quality control thresholds** at **high-motion sites** could improve data reliability.

Data quality for **Diagnostic Accuracy** in deep learning models

End Goal



Acknowledgments

Thank you to my amazing mentors Dr. Margot Wagner and Dr. Alessandra Camassa for the incredible guidance. I would also like to thank Dr. Terrence Sejnowski for the opportunity to conduct research at the Salk Institute for Biological Studies. Lastly thank you to CAMP for support.